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Herbicide-resistant acetyl-CoA carboxylase (ACC) mutant and use thereof

Abstract

A mutant acetyl-CoA carboxylase (ACC) protein, a nucleic acid encoding the mutant ACC protein, and use thereof are provided. Specifically, compared with a parent ACC protein, the mutant ACC protein has mutations at amino acids corresponding to amino acid 1,879 and/or amino acid 2,186 of SEQ ID NO: 1. An ACC-mutated plant shows high herbicide resistance, and thus the present disclosure has very promising application prospects in the cultivation of an herbicide-resistant plant.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure belongs to the fields of biotechnology and crop genetic breeding, and specifically relates to an acetyl-CoA carboxylase (ACC) mutant protein, a nucleic acid encoding the ACC mutant protein, and a method and use for improving the resistance of a plant to a herbicide.

BACKGROUND

(2) *Oryza sativa* is consumed by two-thirds of the world's population and is the main energy source in the diet of at least half of the two-thirds of the world's population. Rice is a low-cost food that is easily and rapidly prepared, which can be eaten together with various dishes.

(3) The use of an herbicide to control weeds or plants in a crop has become an almost universal practice. As an important part of the modern agricultural production system, herbicides are

considered as the most reliable and economical means among farmland weeding techniques. Since 2,4-dichlorophenoxyacetic acid (2,4-D) was first used in the 1940s, the herbicide industry has been developed for more than 60 years, and a large number of selective herbicides have been successfully developed. The research on acetyl-CoA carboxylase (ACC) inhibitors began in the 1970s. ACC herbicides are divided into the following 4 types: aryloxyphenoxypropionates (APPs), cyclohexanedione oximes (CHDs), aryloxyphenyl-cyclohexanediones (APCHDs), and cycloketones (CTRs). ACC herbicides can inhibit the synthesis of fatty acids in gramineous plants with high selectivity, can be transduced in plants, and can control annual or perennial gramineous weeds post-emergence. ACC herbicides have the advantages of high efficiency, low toxicity, long application period, safety to subsequent crops, and the like, and thus occupy an important position in the herbicide market.

(4) ACC is an important target of chemical herbicides, which is a biotin-containing enzyme discovered in 1958. ACC catalyzes the carboxylation of acetyl-CoA in an organism to form malonyl-CoA, which provides a substrate for the synthesis of fatty acids and many secondary metabolites. ACC is a key enzyme or a rate-limiting enzyme for the biosynthesis of fatty acids. The carboxylase involves a two-step reversible reaction, including adenosine triphosphate (ATP)-dependent carboxylation of a biotin group on a substrate domain through biotin-carboxylase enzymatic activity, and transfer of carboxyl from biotin by carboxyltransferase to obtain an acetyl-CoA substrate. ACC is a key enzyme for the biosynthesis of fatty acids in a plant, which occurs in chloroplasts and mitochondria. Moreover, ACC also plays a role in the formation of long-chain fatty acids and flavonoids and the malonylation in cytoplasm.

SUMMARY

(5) The present disclosure is intended to provide a mutant ACC protein capable of imparting herbicide resistance to a plant or a polynucleotide encoding the same, and use thereof.

(6) In the present disclosure, ACC refers to acetyl CoA carboxylase.

(7) Mutant ACC

(8) In an aspect, the present disclosure provides a mutant ACC, and compared to an amino acid sequence of a parent ACC, the mutant ACC has mutation at amino acid corresponding to amino acid 1,879 and/or amino acid 2,186 of an amino acid sequence shown in SEQ ID NO: 1.

(9) In an embodiment, the amino acid 1,879 of the parent ACC may be isoleucine (I), and the amino acid 2,186 may be cysteine (C).

(10) In an embodiment, the isoleucine (I) at position 1,879 may be mutated into an amino acid other than isoleucine (I), and the amino acid other than isoleucine (I) may be one or more selected from the group consisting of alanine (A), valine (V), glycine (G), leucine (L), glutamine (Q), phenylalanine (F), tryptophan (W), tyrosine (Y), aspartic acid (D), asparagine (N), glutamic acid (E), lysine (K), methionine (M), serine (S), threonine (T), cysteine (C), proline (P), histidine (H), and arginine (R).

(11) In a preferred embodiment, the isoleucine (I) at position 1,879 may be mutated into valine (V).

(12) In an embodiment, the cysteine (C) at position 2,186 may be mutated into an amino acid other than cysteine (C), and the amino acid other than cysteine (C) may be one or more selected from the group consisting of alanine (A), valine (V), glycine (G), leucine (L), isoleucine (I), phenylalanine (F), tryptophan (W), tyrosine (Y), aspartic acid (D), asparagine (N), lysine (K), glutamine (Q), methionine (M), serine (S), threonine (T), glutamic acid (E), proline (P), histidine (H), and arginine (R).

(13) In a preferred embodiment, the cysteine (C) at position 2,186 may be mutated into arginine (R).

(14) In an embodiment, the mutation may be selected from the group consisting of I1879V, C2186R, and a combination thereof.

(15) In an embodiment, the parent ACC may be derived from any plant.

(16) In an embodiment, the parent ACC may be derived from one or more selected from the group

consisting of a gramineous plant, a leguminous plant, a chenopodiaceous plant, and a cruciferous plant.

(17) In an embodiment, the parent ACC may be derived from one or more selected from the group consisting of *Arabidopsis thaliana* (*A. thaliana*), *Oryza sativa*, *Nicotiana tabacum*, *Zea mays*, *Sorghum bicolor*, *Hordeum vulgare*, *Triticum aestivum*, *Setaria italica*, *Glycine max*, *Lycopersicon esculentum*, *Solanum tuberosum*, *Chenopodium quinoa*, *Lactuca sativa*, *Brassica napus*, *Brassica pekinensis*, and *Fragaria ananassa*.

(18) In a preferred embodiment, the parent ACC of the present disclosure may be derived from *Oryza* L., especially *Oryza sativa*.

(19) In an embodiment, the parent ACC may have ACC activity, and an amino acid sequence of the parent ACC may have at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% sequence identity with the amino acid sequence shown in SEQ ID NO: 1.

(20) In a preferred embodiment, the amino acid sequence of the parent ACC may have the sequence shown in SEQ ID NO: 1, or the amino acid sequence of the parent ACC may be shown in SEQ ID NO: 1.

(21) In an embodiment, the mutant ACC may have at least 60%, preferably at least 70%, preferably at least 80%, and preferably at least 90% (such as 95%, 97%, or 99%) homology with a sequence shown in any one of SEQ ID NOs.: 2-4.

(22) In an embodiment, the mutant ACC may be a polypeptide with an amino acid sequence shown in any one of SEQ ID NOs.: 2-4, an active fragment thereof, or a conservative variant polypeptide thereof.

(23) In an embodiment, the mutant ACC may have an amino acid sequence shown in any one of SEQ ID NOs.: 2-4.

(24) Fusion Protein

(25) In another aspect, the present disclosure provides a fusion protein including the mutant ACC protein of the present disclosure. Further, the fusion protein may also include: a tag peptide, such as a histidine tag or 6×His; a plastid-targeted peptide, such as a chloroplast-targeted peptide; or a regulatory element, such as a promoter sequence, a terminator sequence, a leader sequence, a polyadenylation sequence, or a marker gene.

(26) Polynucleotide

(27) In another aspect, the present disclosure provides a polynucleotide encoding the mutant ACC protein or an active fragment thereof.

(28) In an embodiment, the polynucleotide may be selected from the group consisting of:

(29) (a) a polynucleotide encoding a protein shown in any one of SEQ ID NOs.: 2-4;

(30) (b) a polynucleotide with a sequence shown in any one of SEQ ID NOs.: 5-7;

(31) (c) a polynucleotide that has a nucleotide sequence of homology $\geq 80\%$ (preferably $\geq 90\%$, more preferably $\geq 95\%$, and most preferably $\geq 98\%$) with a sequence shown in SEQ ID NO: 5 and encodes a protein shown in SEQ ID NO: 2; or, a polynucleotide that has a nucleotide sequence of homology $\geq 80\%$ (preferably $\geq 90\%$, more preferably $\geq 95\%$, and most preferably $\geq 98\%$) with a sequence shown in SEQ ID NO: 6 and encodes a protein shown in SEQ ID NO: 3; or, a polynucleotide that has a nucleotide sequence of homology $\geq 80\%$ (preferably $\geq 90\%$, more preferably $\geq 95\%$, and most preferably $\geq 98\%$) with a sequence shown in SEQ ID NO: 7 and encodes a protein shown in SEQ ID NO: 4; and

(32) (d) a polynucleotide complementary to any one selected from the group consisting of the polynucleotides described in (a) to (c).

(33) In an embodiment, the polynucleotide may be selected from the group consisting of a genomic sequence, a cDNA sequence, an RNA sequence, and a combination thereof.

(34) In an embodiment, the polynucleotide may preferably be single-stranded or double-stranded.

(35) In an embodiment, the polynucleotide may make an auxiliary element selected from the group

consisting of a signal peptide, a secretory peptide, a tag sequence (such as 6His), a nuclear localization signal (NLS), and a combination thereof additionally included at a flank of an open reading frame (ORF) of the mutant protein.

(36) In an embodiment, the polynucleotide may further include a promoter operably linked to the ORF sequence of the mutant polypeptide.

(37) In an embodiment, the promoter may be selected from the group consisting of a constitutive promoter, a tissue-specific promoter, an inducible promoter, and a strong promoter.

(38) Nucleic Acid Construct

(39) In another aspect, the present disclosure provides a nucleic acid construct, including the polynucleotide and a regulatory element operably linked thereto.

(40) In an embodiment, the regulatory element may be one or more selected from the group consisting of an enhancer, a transposon, a promoter, a terminator, a leader sequence, a polyadenylate sequence, and a marker gene.

(41) Vector

(42) The present disclosure also provides a vector including a nucleic acid sequence encoding the mutant ACC or the fusion protein of the present disclosure. Preferably, the vector may further include an expression regulation element operably linked to the aforementioned nucleic acid sequence.

(43) In an embodiment, the vector may include a cloning vector, an expression vector, a shuttle vector, and an integration vector.

(44) In an embodiment, the vector may be a vector for gene editing of the endogenous ACC gene in a host cell.

(45) In an embodiment, the vector may include a polynucleotide encoding a polypeptide shown in any one of SEQ ID NOs.: 2-4.

(46) In an embodiment, the expression vector may also include at least one replication origin to realize self-replication.

(47) In an embodiment, the vector may be a vector that will be integrated into a genome of a host cell when introduced into the host cell and then replicates together with a chromosome into which the vector is integrated.

(48) The vector can be, for example, a plasmid, a virus, a cosmid, a phage, and the like, which are well known to those skilled in the art.

(49) Preferably, the vector in the present disclosure may be a plasmid.

(50) Edit Vector System

(51) In another aspect, the present disclosure provides an edit vector system that can produce the above-mentioned mutant ACC in a plant. The edit vector system may include one or more vectors, and the one or more vectors may at least include a guide sequence targeting the parent ACC.

(52) The guide sequence may include a part of the nucleotide sequence of the parent ACC, preferably at least 15 bp of the ACC nucleotide sequence and more preferably at least 20 bp of the ACC nucleotide sequence. In an embodiment, the edit vector may further include a gene editing enzyme. The gene editing enzyme may include nucleases suitable for CRISPR, transcription activator-like (TAL) effector nuclease (TALEN), zinc-finger nuclease (ZFN), and other editing tools.

(53) Preferably, the gene editing enzyme may be a Cas protein, also known as a CRISPR enzyme or a Cas effector protein, including but not limited to: a Cas9 protein, a Cas12 protein, a Cas13 protein, a Cas14 protein, a Csm1 protein, and an FDK1 protein.

(54) Preferably, the Cas protein may be operably linked to a first regulatory element.

(55) In an embodiment, the gene editing enzyme may be a Cas9 protein, and the vector may further include a Scaffold sequence that can specifically bind to the Cas9 protein. The Scaffold sequence and the guide sequence are operably linked to form a gRNA. Preferably, the gRNA may be operably linked to a second regulatory element.

(56) In other embodiments, the gene editing enzyme may be a Cas12 protein, such as Cas12a, Cas12b, and Cas12i, and the vector may further include a direct repeat that specifically binds to the Cas12 protein. The direct repeat and the guide sequence are operably linked to form a gRNA. Preferably, the gRNA may be operably linked to the second regulatory element.

(57) The above-mentioned regulatory elements may include a promoter, a terminator sequence, a leader sequence, a polyadenylation sequence, a signal peptide coding region, a marker gene, an enhancer, an internal ribosome entry site (IRES), and other expression control elements (for example, a transcriptional termination signal, such as a polyadenylation signal and a poly-U sequence).

(58) Preferably, the edit vector system may further include a base editing element, which is adenine deaminase and/or cytosine deaminase.

(59) In an embodiment, the edit vector may also include a resistance gene to facilitate screening; and the resistance gene may include hyg, bar, kana, rif, spec, and amp, which are well known to those skilled in the art.

(60) Preferably, the guide sequence of gRNA may be A-ACC1879:

GAGAATATACATGGAAGTGC; A-ACC2186: ATAGCACTCAATGCGGTCTG; or a combination thereof.

(61) Preferably, the Cas protein may be nCas9 or another Cas9 protein with nick activity. The “n” stands for nick, that is, the “n” indicates a Cas protein with only single-strand cleavage activity.

(62) Host Cell

(63) In another aspect, the present disclosure provides a host cell, where the host cell includes one or more selected from the group consisting of the mutant ACC, the gene encoding the mutant ACC, the fusion protein, the vector, and the nucleic acid construct; or, the polynucleotide is integrated into a genome of the host cell.

(64) In an embodiment, the host cell may be a prokaryotic cell, such as *Escherichia coli* (*E. coli*).

(65) In an embodiment, the host cell may be a plant cell, and the plant may include an angiosperm and a gymnosperm.

(66) In an embodiment, the plant may include a monocotyledonous plant and a dicotyledonous plant.

(67) In an embodiment, the plant may include an herbaceous plant and a woody plant.

(68) In an embodiment, the plant may include *A. thaliana*, *Nicotiana tabacum*, *Oryza sativa*, *Zea mays*, *Sorghum bicolor*, *Hordeum vulgare*, *Triticum aestivum*, *Setaria italica*, *Glycine max*, *Lycopersicon esculentum*, *Solanum tuberosum*, *Chenopodium quinoa*, *Lactuca sativa*, *Brassica napus*, *Brassica pekinensis*, and *Fragaria ananassa*.

(69) Resistant Plant

(70) In another aspect, the present disclosure provides an herbicide-resistant plant, where the herbicide-resistant plant includes one or more selected from the group consisting of the mutant ACC, the polynucleotide encoding the mutant ACC, the fusion protein, the vector, and the nucleic acid construct; or, the polynucleotide is integrated into a genome of the plant.

(71) The term “substantially identical” means that two amino acid sequences have 85%, 90%, 95%, 96%, 97%, 98%, or 99% identify.

(72) Method for Preparing the Mutant Polypeptide

(73) In another aspect, the present disclosure provides a method for preparing the mutant ACC polypeptide or an active fragment thereof, including the following step:

(74) (a) under conditions suitable for expression, cultivating a host cell including the mutant ACC polypeptide to express the mutant ACC polypeptide; and preferably, the method may further include:

(75) (b) isolating the mutant ACC polypeptide.

(76) Method for Acquiring the Herbicide-Resistant Plant

(77) In another aspect, the present disclosure provides a plant cell, a plant seed, a plant tissue, a

plant part, or a plant with herbicide resistance, where the plant cell, the plant tissue, the plant seed, the plant part, or the plant includes the mutant ACC polypeptide or a polynucleotide sequence encoding the same.

(78) In another aspect, the present disclosure provides a method for acquiring or preparing a plant cell, a plant seed, a plant tissue, a plant part, or a plant with herbicide resistance, including: introducing the mutant ACC polypeptide or a polynucleotide sequence encoding the same into the plant cell, the plant seed, the plant tissue, the plant part, or the plant.

(79) In an embodiment, the introducing the ACC mutant polypeptide of the present disclosure may include the following step: allowing the ACC mutant polypeptide to express in the plant cell, the plant seed, the plant tissue, the plant part, or the plant. For example, the mutant polypeptide is expressed by an expression vector, or the mutant polypeptide is integrated into a plant genome for expression.

(80) In a preferred embodiment, the above method may include the following steps:

(81) (1) providing *Agrobacterium tumefaciens* (*A. tumefaciens*) carrying an expression vector, where the expression vector includes a DNA coding sequence of the mutant ACC polypeptide or an active fragment thereof;

(82) (2) contacting the plant cell, the plant tissue, or the plant part with the *A. tumefaciens* in step (1), such that the DNA coding sequence of the mutant ACC polypeptide or the active fragment thereof is transformed into the plant cell and integrated on a chromosome of the plant cell; and

(83) (3) screening out a plant cell transformed with the DNA coding sequence of the mutant ACC polypeptide or the active fragment thereof.

(84) In an embodiment, the introducing the mutant ACC polypeptide may include the following step: allowing endogenous ACC of the plant to mutate to introduce the mutant polypeptide. Preferably, the mutant polypeptide may be introduced by gene editing.

(85) In another preferred embodiment, the method may include: allowing an endogenous ACC coding sequence in the plant cell, the plant seed, the plant tissue, or the plant part to mutate at positions corresponding to amino acid 1,879 and/or amino acid 2,186 of SEQ ID NO: 1.

(86) In another preferred embodiment, the method may include following steps:

(87) (1) introducing the aforementioned edit vector system into the plant cell, the plant seed, the plant tissue, or the plant part; and

(88) (2) allowing a gene editing tool to act on an endogenous ACC coding sequence, such that the endogenous ACC coding sequence mutates at positions corresponding to amino acid 1,879 and/or amino acid 2,186 of the amino acid sequence shown in SEQ ID NO: 1.

(89) Further, the above method may also include the following step: screening a mutant plant cell, plant tissue, or plant part, and optionally, isolating the gene editing tool.

(90) In another preferred embodiment, the gene editing tool may include CRISPR, TALEN, and ZFN.

(91) In another preferred embodiment, the plant may include an angiosperm and a gymnosperm.

(92) In another preferred embodiment, the plant may include a monocotyledonous plant and a dicotyledonous plant.

(93) In another preferred embodiment, the plant may include an herbaceous plant and a woody plant.

(94) In another preferred embodiment, the plant may include *A. thaliana*, *Nicotiana tabacum*, *Oryza sativa*, *Zea mays*, *Sorghum bicolor*, *Hordeum vulgare*, *Triticum aestivum*, *Setaria italica*, *Glycine max*, *Lycopersicon esculentum*, *Solanum tuberosum*, *Chenopodium quinoa*, *Lactuca sativa*, *Brassica napus*, *Brassica pekinensis*, and *Fragaria ananassa*.

(95) In an embodiment, compared with the maximum tolerated herbicide concentration of the parent plant, the maximum tolerated herbicide concentration of the plant with the mutant polypeptide may be increased by at least 1 to 4 times, such as 2, 3, or 4 times. When an herbicide is applied, the plant with the mutant polypeptide of the present disclosure can tolerate the herbicide at

a concentration least 2 to 4 times a recommended concentration of the herbicide, while the parent plant cannot tolerate the herbicide at this concentration.

(96) Method for Controlling Weeds

(97) In another aspect, the present disclosure also provides a method for controlling the growth of weeds near a plant, including:

(98) a) providing the above-mentioned herbicide-resistant plant; and

(99) b) applying an effective amount of the herbicide to the plant and weeds nearby to control the weeds.

(100) The plant may preferably be *Oryza sativa*.

(101) The herbicide may include one or more selected from the group consisting of APP, CHD, APCHD, and CTR. Preferably, the herbicide may include one or more selected from the group consisting of alloxydim, butroxydim, clethodim, cyclohexenone, cycloxydim, sethoxydim, tepraloxym, tralkoxydim, phentodim, chloroazifop-propynyl, clodinafop-propargyl, clofop, diclofop-methyl, fenoxaprop-ethyl, fenoxaprop-p-ethyl, fenthiaprop-ethyl, fluazifop-butyl, fluazifop-p-butyl, haloxyfop-methyl, haloxyfop-p-methyl, clomazone, clefoxidim, propaquizafop, quizalofop, quizalofop-p, quizalofop-p-ethyl, quizalofop-p-tefuryl, trifluorophenoxy propionic acid, fenoxaprop, and a salt or ester thereof.

(102) Preferably, the herbicide may be one or more selected from the group consisting of sethoxydim, phentodim, and clethodim.

(103) Use

(104) In another aspect, the present disclosure provides use of the mutant ACC, the gene encoding the mutant ACC, the fusion protein, the vector, or the nucleic acid construct in a reagent or a kit for preparing an herbicide-resistant plant.

(105) In another aspect, the present disclosure provides use of the mutant ACC, the gene encoding the mutant ACC, the fusion protein, the vector, or the nucleic acid construct for controlling weeds.

(106) In another aspect, the present disclosure provides use of the mutant ACC, the gene encoding the mutant ACC, the fusion protein, the vector, or the nucleic acid construct in the preparation of an herbicide-resistant plant.

(107) Herbicide

(108) In an embodiment, the herbicide of the present disclosure may be an ACC-inhibiting herbicide, and the ACC-inhibiting herbicide includes, but is not limited to, one or more selected from the group consisting of APP, CHD, APCHD, and CTR.

(109) Preferably, the herbicide of the present disclosure includes, but is not limited to, one or more selected from the group consisting of alloxydim, butroxydim, clethodim, cyclohexenone, cycloxydim, sethoxydim, tepraloxym, tralkoxydim, phentodim, chloroazifop-propynyl, clodinafop-propargyl, clofop, diclofop-methyl, fenoxaprop-ethyl, fenoxaprop-p-ethyl, fenthiaprop-ethyl, fluazifop-butyl, fluazifop-p-butyl, haloxyfop-methyl, haloxyfop-p-methyl, clomazone, clefoxidim, propaquizafop, quizalofop, quizalofop-p, quizalofop-p-ethyl, quizalofop-p-tefuryl, trifluorophenoxy propionic acid, fenoxaprop, and a salt or ester thereof. Preferably, the herbicide may be one or more selected from the group consisting of phentodim, sethoxydim, and clethodim.

General Definitions

(110) Unless defined otherwise, the technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art.

(111) The terms “polynucleotide”, “nucleotide sequence”, “nucleic acid sequence”, “nucleic acid molecule”, and “nucleic acid” may be used interchangeably and include DNA, RNA, or a hybrid thereof, which may be double-stranded or single-stranded.

(112) The term “homology” or “identity” used refers to sequence matching between two polypeptides or between two nucleic acids. Therefore, the composition and method of the present disclosure also include homologues of the nucleotide sequence and the polypeptide sequence (such as SEQ ID NOs: 1-7) of the present disclosure. “Homology” can be calculated by a known method

including but not limited to: Computational Molecular Biology (edited by Lesk, A. M.), Oxford University Press, New York (1988); Biocomputing: Informatics and Genome Projects (edited by Smith, D. W.), Academic Press, New York (1993); Computer Analysis of Sequence Data, Part I (edited by Griffin, A. M. and Griffin, H. G.), Humana Press, New Jersey (1994); Sequence Analysis in Molecular Biology (edited by von Heinje, G.), Academic Press (1987); and Sequence Analysis Primer (edited by Gribskov, M. and Devereux, J.), Stockton Press, New York (1991).

(113) A specific amino acid position (number) in the protein of the present disclosure is determined by aligning an amino acid sequence of the target protein with SEQ ID NO: 1 using a standard sequence alignment tool. For example, the Smith-Waterman algorithm or CLUSTALW2 algorithm is used to align two sequences, and the sequences are considered aligned when an alignment score is the highest. The alignment score can be calculated according to the method described in Wilbur, W. J. and Lipman, D. J. (1983) Rapid similarity searches of nucleic acid and protein data banks. Proc. Natl. Acad. Sci. USA, 80: 726-730. In the ClustalW2 (1.82) algorithm, default parameters may be preferably used: protein gap opening penalty=10.0; protein gap extension penalty=0.2; protein matrix=Gonnet; protein/DNA end gap=-1; and protein/DNAGAPDIST=4. The AlignX program (a part of the vectorNTI group) may preferably be used to align an amino acid sequence of a protein with SEQ ID No: 1 using default parameters suitable for multiple alignments (gap opening penalty: 10 og; and gap extension penalty: 0.05) to determine a position of a specific amino acid within the protein.

(114) The term “encoding” refers to an inherent characteristic of a specific nucleotide sequence in a polynucleotide, such as a gene cDNA or mRNA, which serves as a template for the synthesis of a defined nucleotide sequence (namely, rRNA, tRNA, and mRNA) or a defined amino acid sequence and the synthesis of other polymers and macromolecules in a biological process of a biological characteristic thereof. Therefore, if the transcription and translation of mRNA corresponding to a gene produces a protein in a cell or another biological system, the gene encodes the protein.

(115) The term “amino acid” refers to a carboxylic acid with amino. Various proteins in organisms are composed of 20 essential amino acids.

(116) The terms “protein”, “polypeptide”, and “peptide” can be used interchangeably in the present disclosure and refer to a polymer of amino acid residues, including a polymer in which one or more amino acid residues are a chemical analogue of a natural amino acid residue. The protein and polypeptide of the present disclosure can be produced through recombination or chemical synthesis.

(117) The term “mutant protein” refers to a protein that is obtained through substitution, insertion, deletion, and/or addition of one or more amino acid residues based on an amino acid sequence of a parent protein.

(118) The term “AxxB” means that amino acid A at position xx is changed into amino acid B. For example, I1879V means that I at position 1,879 is changed into V, C2186R means that C at position 2,186 is changed into R, and so on. For double or multiple mutations, mutations are separated by a “/”. For example, I1879V/C2186R indicates that, relative to the amino acid sequence of SEQ ID NO: 1, I at position 1,879 is substituted by V and C at position 2,186 is substituted by R, where both mutations are present in the specific mutant ACC protein.

(119) As used herein, the term “regulatory element” is intended to include a promoter, a terminator sequence, a leader sequence, a polyadenylation sequence, a signal peptide coding region, a marker gene, an enhancer, an IRES, and other expression control elements (for example, a transcriptional termination signal, such as a polyadenylation signal and a poly-U sequence), and the detailed description can be seen in Goeddel, “*GENE EXPRESSION TECHNOLOGY:METHODS IN ENZYMOLOGY*” 185, Academic Press, San Diego, Calif. (1990). In some cases, the regulatory element includes sequences that guide the constitutive expression of a nucleotide sequence in many types of host cells and sequences that guide the expression of the nucleotide sequence only in some host cells (such as a tissue-specific regulatory sequence). A tissue-specific promoter can mainly

guide the expression in a desired tissue of interest, such as muscles, neurons, bones, skin, blood, specific organs (such as liver and pancreas), or specific cell types (such as lymphocytes). In some cases, a regulatory element can also guide the expression in a time-dependent manner (such as in a cell cycle-dependent or developmental stage-dependent manner), which may be or may not be tissue or cell type-specific. In some cases, the term “regulatory element” covers enhancer elements, such as WPRE; CMV enhancer; R-U5' fragment in LTR of HTLV-I ((Mol. Cell. Biol., Vol 8 (1): 466-472, 1988); SV40 enhancer; and an intron sequence between exons 2 and 3 of rabbit β -globin (Proc. Natl. Acad. Sci. USA., Vol. 78 (3): 1527-31, 1981).

(120) As used herein, the term “promoter” has the meaning well known to those skilled in the art, which refers to a non-coding nucleotide sequence located upstream of a gene and capable of promoting the expression of a downstream gene. A constitutive promoter is a nucleotide sequence that will result in the generation of a gene product in a cell under most or all physiological conditions of the cell after the promoter is operably linked to a polynucleotide encoding or defining the gene product. An inducible promoter is a nucleotide sequence that will cause the generation of a gene product in a cell only when there is an inducer corresponding to the promoter in the cell after the promoter is operably linked to a polynucleotide encoding or defining the gene product. A tissue-specific promoter is a nucleotide sequence that will cause the generation of a gene product in a cell basically only when the cell is a cell of the tissue type corresponding to the promoter after the promoter is operably linked to a polynucleotide encoding or defining a gene product.

(121) “NLS” (Nuclear Localization Signal) is an amino acid sequence that tags a protein for import into the nucleus through nuclear transport, that is, a protein with NLS is transported to the nucleus. Typically, NLS may include positively charged Lys or Arg residues that are exposed on the surface of a protein. Exemplary NLS includes, but is not limited to, SV40 large T antigen, EGL-13, c-Myc, and TUS protein.

(122) As used herein, the term “operably linked” means that a nucleotide sequence of interest is linked to one or more regulatory elements in a manner that allows the expression of the nucleotide sequence (for example, in an in vitro transcription/translation system or in a host cell when the vector is introduced into the host cell).

(123) The term “vector” refers to an element that is allowed to be integrated into a genome of a host cell or to self-replicate within a cell independently of its genome. The vector may include any elements that guarantee the self-replication. The vector usually carries a gene that is not a part of the central metabolism of a cell and is usually in the form of double-stranded DNA. The selection of a vector generally depends on the compatibility of the vector with a host cell into which the vector is to be introduced. When a vector needs to be used, the selection of the vector depends on a method for transforming a host cell well known to those skilled in the art. For example, a plasmid vector can be used.

(124) The term “ACC herbicide” refers to an herbicide that can inhibit the synthesis of fatty acids in gramineous plants with high selectivity, can be transduced in plants, and can control annual or perennial gramineous weeds post-emergence. The ACC herbicides have the advantages of high efficiency, low toxicity, long application period, safety to subsequent crops, and the like, and thus occupy an important position among herbicides.

(125) The ACC herbicides can be divided into the following 4 types: APPs, CHDs, APCHDs, and CTRs, which are shown as follows.

(126) ##STR00001##

(127) The herbicide includes, but is not limited to, alloxymid, butoxydim, clethodim, cyclohexenone, cycloxydim, sethoxydim, tepraloxymid, tralkoxydim, chloroazifop-propynyl, clodinafop-propargyl, clofop, diclofop-methyl, fenoxaprop-ethyl, fenoxaprop-p-ethyl, fenthiaprop-ethyl, fluazifop-butyl, fluazifop-p-butyl, haloxyfop-methyl, haloxyfop-p-methyl, clomazone, clefloxidim, propaquizafop, quizalofop, quizalofop-p, quizalofop-p-ethyl, quizalofop-p-tefuryl, trifluorophenoxy propionic acid, and fenoxaprop.

(128) The “herbicide resistance” refers to the genetic ability of a plant to survive and reproduce after the plant is exposed to an herbicide at a dosage that is usually lethal to the wild-type plant. In plants, the herbicide resistance can be naturally produced or induced by a technique such as genetic engineering or screening a variant produced by tissue culture or mutagenesis. Unless otherwise stated, the herbicide resistance is heritable and allows a plant to grow and reproduce in the presence of a typical effective herbicidal treatment for a given plant by the herbicide, as suggested by the current version of “Herbicide Manual” when the present disclosure is filed. As recognized by those skilled in the art, a plant can still be considered “resistant”, even if a specified degree of plant damage due to herbicide exposure is obvious. As used herein, the term “tolerance” includes “resistance” or “resistant plant” as defined herein, and the improved ability of a specific plant to tolerate various degrees of damage induced by an herbicide (typically ethyl) at the same herbicide dosage as the wild-type plant of the same genotype.

(129) The term “parent ACC polypeptide” refers to a polypeptide derived from the ACC mutant polypeptide. In a preferred embodiment, the parent ACC polypeptide is a protein (polypeptide) that can be found in nature or is encoded by a nucleic acid that can be found in nature, where nucleotides of the nucleic acid can be obtained through genetic engineering such as genome sequencing and polymerase chain reaction (PCR), and an amino acid sequence of the protein can be deduced from the nucleotide sequence. An amino acid sequence of the wild-type ACC polypeptide is shown in SEQ ID NO: 1, for example. In some embodiments, the parent ACC polypeptide may be obtained by changing one or more amino acid residues of the wild-type ACC polypeptide without affecting the enzymatic activity.

(130) The terms “mutant ACC protein”, “mutant ACC”, “mutant ACC enzyme”, “mutant protein”, “mutant polypeptide”, “polypeptide of the present disclosure”, “protein of the present disclosure”, and the like can be used interchangeably. Preferably, the mutant protein may be obtained through mutation of amino acid 1,879 and/or amino acid 2,186 of the sequence shown in SEQ ID NO: 1.

(131) The term “host organism” should be understood as any unicellular or multicellular organism into which a nucleic acid encoding the mutant ACC protein can be introduced, including, for example, bacteria such as *E. coli*, fungi such as yeast (such as *Saccharomyces cerevisiae* (*S. cerevisiae*)), molds (such as *Aspergillus*), plant cells, and plants.

(132) The term “plant” should be understood as any differentiated multicellular organism capable of photosynthesis, including: crop plants at a mature or developmental stage, especially monocotyledonous or dicotyledonous plants; vegetable crops including artichoke, turnip cabbage, arugula, leek, asparagus, lettuce (such as head lettuce, leaf lettuce, and romaine lettuce), bok choy, malanga, melons (such as cantaloupe, watermelon, crenshaw melon, honeydew melon, and Roman cantaloupe), rape crops (such as Brussels sprout, cabbage, cauliflower, broccoli, borecole, kale, Chinese cabbage, and bok choy), cardoon, carrot, napa, okra, onion, celery, parsley, chickpea, parsnip, chicory, pepper, *Solanum tuberosum*, gourd (such as marrow squash, cucumber, zucchini, cushaw, and pumpkin), radish, dried ball onion, rutabaga, purple eggplant (also known as eggplant), salsify, lettuce, shallot, endive, garlic, spinach, green onion, cushaw, greens, beets (sugar beets and fodder beets), sweet potato, Swiss chard, wasabi, tomato, turnip, and spices; fruits and/or vine crops such as apple, apricot, cherry, nectarine, peach, pear, plum, prune, cherry, quince, almond, chestnut, hazelnut, pecan, pistachio, walnut, citrus, blueberry, boysenberry, cranberry, currant, loganberry, raspberry, strawberry, blackberry, grape, avocado, banana, kiwi, persimmon, pomegranate, pineapple, tropical fruit, pome, melon, mango, papaya, and lychee; field crops, such as clover, alfalfa, evening primrose, meadowfoam, corn/maize (forage corn, sweet corn, and popcorn), lupulus, jojoba, peanut, rice, safflower, small grain crops (*Hordeum vulgare*, oat, rye, *Triticum aestivum*, and the like), *Sorghum bicolor*, *Nicotiana tabacum*, kapok, legumes (beans, lentil, pea, and *Glycine max*), oil plants (canola, leaf mustard, poppy, olive, sunflower, coconut, castor oil plant, cocoa bean, and groundnut), *Arabidopsis*, fiber plants (cotton, flax, hemp, and jute), *Lauraceae* (cinnamon or camphor), or a plant such as coffee, sugar cane, tea, and natural

rubber plants; and/or bedding plants such as a flowering plant, cactus, a succulent plant, and/or an ornamental plant, and trees such as forests (broad-leaved and evergreen trees, such as conifers), fruit trees, ornamental trees, nut-bearing trees, shrubs, and other seedlings.

(133) The term “plant tissue” or “plant part” includes a plant cell, a protoplast, a plant tissue culture, a plant callus, a plant piece, a plant embryo, a pollen, an ovule, a seed, a leaf, a stem, a flower, a branch, a seedling, a fruit, a nucleus, a spike, a root, a root tip, an anther, and the like.

(134) The term “plant cell” should be understood as any cell derived or found in a plant, which is capable of forming, for example, undifferentiated tissues such as calli, differentiated tissues such as embryos, constituent parts of a plant, plants, or seeds.

(135) The gene editing technology includes CRISPR technology, TALEN technology, and ZFN technology. “CRISPR” refers to clustered regularly interspaced short palindromic repeat, which comes from the immune system of microorganisms. A gene editing tool includes gRNA and Cas protein (such as Cas9, Cpf1, and Cas12b). The gene editing tool of TALEN refers to a restriction enzyme that can cleave a specific DNA sequence, which includes a TAL effector DNA binding domain and a DNA cleavage domain. The gene editing tool of ZFN refers to a restriction enzyme that can cleave a specific DNA sequence, which includes a zinc-finger DNA binding domain and a DNA cleavage domain. It is well known to those skilled in the art that an intracellular genome can be edited by constructing nucleotides encoding a gene editing tool and other regulatory elements into a suitable vector and then transforming the vector into a cell; and a type of the editing includes gene knockout, insertion, and base editing.

(136) As used herein, the term “gene editing enzyme” refers to a nuclease suitable for CRISPR, TALEN, ZFN, and other editing tools. Preferably, the gene editing enzyme may be a CRISPR enzyme, also known as a Cas protein, including but not limited to: a Cas9 protein, a Cas12 protein, a Cas13 protein, a Cas14 protein, a Csm1 protein, and an FDK1 protein. The Cas protein refers to a protein family. Cas proteins have different structures according to different sources, such as SpCas9 derived from *Streptococcus pyogenes* (*S. pyogenes*) and SaCas9 derived from *Staphylococcus aureus* (*S. aureus*). Cas proteins can also be classified based on structural characteristics (such as domains), for example, the Cas12 family may include Cas12a (also known as Cpf1), Cas12b, Cas12c, Cas12i, and the like. The Cas protein may be double-stranded or single-stranded or may have no cleavage activity. The Cas protein of the present disclosure may be wild-type or a mutant thereof. A mutation type of the mutant may include amino acid substitution, substitution, or deletion. The mutant may or may not change the enzyme cleavage activity of the Cas protein. Preferably, the Cas protein of the present disclosure may only have single-strand cleavage activity or may not have cleavage activity, which may be a mutant of the wild-type Cas protein. Preferably, the Cas protein of the present disclosure may be Cas9, Cas12, Cas13, or Cas14 with single-strand cleavage activity. In a preferred embodiment, the Cas9 protein of the present disclosure may include SpCas9n(D10A), nSpCas9NG, SaCas9n, ScCas9n, and XCas9n, where “n” stands for nick, indicating a Cas protein with only single-strand cleavage activity. The mutation of a known Cas protein to obtain a Cas protein with single-strand cleavage activity or without cleavage activity is a conventional technique in the art. Those skilled in the art know that a variety of Cas proteins with nucleic acid cleavage activity have been reported in the art; and the known proteins or modified variants can all achieve the functions of the present disclosure, which are incorporated into the protection scope of the present disclosure by reference.

(137) For example, it is clear to those skilled in the art that a structure of a protein can be changed without adversely affecting the activity and functionality of the protein. For example, one or more conservative amino acid substitutions can be introduced into an amino acid sequence of a protein without adversely affecting the activity and/or three-dimensional (3D) configuration of the protein molecule. Those skilled in the art are aware of examples and implementations of the conservative amino acid substitutions. Specifically, an amino acid residue can be substituted by another amino acid residue that belongs to the same group as the amino acid residue to be substituted. That is, a

nonpolar amino acid residue can be substituted by another nonpolar amino acid residue; an uncharged polar amino acid residue can be substituted by another uncharged polar amino acid residue; a basic amino acid residue can be substituted by another basic amino acid residue; and an acidic amino acid residue can be substituted by another acidic amino acid residue. Such substituted amino acid residues may be or may not be encoded by genetic codes. As long as a substitution does not damage the biological activity of a protein, a conservative substitution in which an amino acid is substituted by another amino acid belonging to the same group falls within the scope of the present disclosure. Therefore, in addition to the above-mentioned mutations, the mutant GBSS1 protein of the present disclosure may include one or more other mutations such as conservative substitutions in the amino acid sequence. In addition, the present disclosure covers mutant ACC proteins with one or more other non-conservative substitutions, as long as the non-conservative substitutions do not significantly affect the desired function and biological activity of the protein of the present disclosure.

(138) As well known in the art, one or more amino acid residues can be deleted from the N and/or C terminus of a protein while still retaining its functional activity. Therefore, in another aspect, the present disclosure also relates to a fragment that is obtained through deletion of one or more amino acid residues from the N-terminus and/or C-terminus of the mutant ACC protein and retains the required functional activity (such as an amino acid fragment with the mutation site of the present disclosure), which is also within the scope of the present disclosure and is called a biologically-active fragment. In the present disclosure, the “biologically-active fragment” refers to a part of the mutant ACC protein of the present disclosure, which retains the biological activity of the mutant ACC protein of the present disclosure. For example, the biologically-active fragment of the mutant ACC protein may be obtained after one or more (for example, 1-50, 1-25, 1-10, or 1-5, such as 1, 2, 3, 4, or 5) amino acid residues are deleted from the N-terminus and/or C-terminus of the protein, which still retains the biological activity of the full-length protein.

(139) In addition, the mutant protein of the present disclosure can be modified. Modified (usually without changing the primary structure) forms may include: chemically derived forms of the mutant protein in vivo or in vitro, such as acetylated or carboxylated form. The modification may also include glycosylation, such as glycosylation modification during the synthesis and processing or further processing of the mutant protein to produce a mutant protein. The modification can be accomplished by exposing the mutant protein to a glycosylase (such as a mammalian glycosylase or deglycosylase). The modified forms may also include sequences with phosphorylated amino acid residues (such as phosphotyrosine, phosphoserine, and phosphothreonine). The mutant protein can also be modified to increase its proteolysis resistance or optimize its solubility.

(140) The present disclosure also provides a polynucleotide encoding the mutant ACC polypeptide, which may also include additional coding and/or non-coding sequences. Preferably, the mutant ACC polypeptide may be shown in SEQ ID NOs: 2-4. Those skilled in the art are well aware that, due to the degeneracy of genetic codes, there are many different nucleic acid sequences that can encode the amino acid sequence disclosed herein. Producing other nucleic acid sequences encoding the same protein is within the competence scope of those of ordinary skill in the art, and thus the present disclosure covers nucleic acid sequences encoding the same amino acid sequence due to the degeneracy of genetic codes. For example, in order to achieve the high expression of a heterologous gene in a target host organism such as a plant, the gene can be optimized using a codon preferred by the host organism to allow better expression.

(141) The full-length sequence of the polynucleotide of the present disclosure can usually be obtained through PCR amplification, recombination, or artificial synthesis. For the PCR amplification, primers can be designed according to the relevant nucleotide sequence disclosed in the present disclosure, especially the ORF sequence, and a commercially available cDNA library or a cDNA library prepared by a conventional method known to those skilled in the art can be used as a template to amplify the relevant sequence. When the sequence is long, it is often necessary to

conduct two or more PCR amplifications, and then amplified fragments are spliced together in a correct order. The obtained nucleotide sequence can be cloned into a vector and then transformed into a cell, and then a large number of related sequences can be isolated from proliferated host cells by a conventional method. The mutation site of the present disclosure can also be introduced through artificial synthesis.

(142) One or more copies of the polynucleotide of the present disclosure can be inserted into a host cell to increase a yield of a gene product. The copy number of the polynucleotide can be increased by integrating at least one additional copy of the sequence into a host cell genome or by integrating an amplifiable selectable marker gene with the polynucleotide, where in the latter case, a cell with the amplified copy of the selectable marker gene and the resulting additional copy of the polynucleotide can be selected by artificially cultivating the cell in the presence of a suitable selectable agent.

(143) Methods well known to those skilled in the art can be used to construct a vector that includes a DNA sequence encoding the ACC mutant polypeptide and an appropriate transcription/translation control signal. The methods include in vitro recombinant DNA technology, DNA synthesis technology, and in vivo recombination technology. The DNA sequence can be effectively linked to an appropriate promoter in a vector to guide mRNA synthesis. The vector may also include a ribosome binding site (RBS) for translation initiation and a transcription terminator.

(144) The vectors applicable in the present disclosure may include commercially available plasmids, such as but not limited to: pBR322 (ATCC37017), pKK223-3 (Pharmacia Fine Chemicals, Uppsala, Sweden), GEM1 (Promega Biotec, Madison, Wis., USA), pQE70, pQE60, pQE-9 (Qiagen), pD10, psiX174pBluescript II KS, pNH8A, pNH16a, pNH18A, pNH46A (Stratagene), ptrc99a, pKK223-3, pKK233-3, pDR540, pRIT5 (Pharmacia), pKK232-8, pCM7, pSV2CAT, pOG44, pXT1, pSG (Stratagene), pSVK3, pBPV, pMSG, and pSVL (Pharmacia).

(145) The present disclosure also provides a host cell carrying a nucleic acid sequence, a nucleic acid construct, or an expression vector encoding the ACC mutant polypeptide of the present disclosure. A vector carrying a nucleic acid encoding the protein of the present disclosure is introduced into a host cell, such that the vector exists as a part of a chromosomal integration or exists as a self-replicating extrachromosomal vector described early, or the vector can achieve gene editing on the endogenous ACC gene of the host cell. The host cell may be any host cell familiar to those skilled in the art, including a prokaryotic cell and a eukaryotic cell.

(146) The nucleic acid sequence, nucleic acid construct, or expression vector of the present disclosure can be introduced into a host cell through a variety of techniques, including transformation, transfection, transduction, viral infection, gene gun or Ti-plasmid-mediated gene delivery, calcium phosphate transfection, DEAE-dextran-mediated transfection, lipofection, electroporation, or the like.

(147) In the production method of the present disclosure, the cells are cultivated on a nutrient medium suitable for the production of the polypeptide by a method well known in the art. If the polypeptide is secreted into the nutrient medium, the polypeptide can be directly recovered from the medium. If the polypeptide is not secreted into the medium, the polypeptide can be recovered from a cell lysate.

(148) As used herein, the terms “gRNA”, “mature crRNA”, and “guide sequence” can be used interchangeably and have the meaning commonly understood by those skilled in the art. Generally, a gRNA can include direct repeats and guide sequences, or may be basically composed of or may be composed of direct repeats and guide sequences (also called spacers in the context of endogenous CRISPR systems).

(149) In some cases, the guide sequence can be any polynucleotide sequence that shows sufficient complementarity with a target sequence to hybridize with the target sequence and guide the specific binding of the CRISPR/Cas complex to the target sequence. In an embodiment, under optimal alignment, a complementarity degree between the guide sequence and a corresponding target

sequence may be at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or at least 99%. Determining the optimal alignment is within the competence of those of ordinary skill in the art. For example, there are published and commercially available alignment algorithms and programs, including but not limited to Smith-Waterman, Bowtie, Geneious, Biopython, and SeqMan in ClustalW and matlab.

(150) TABLE-US-00001 Sequence Listing Sequence No. Type Content SEQ ID NO: 1 Protein Amino acid sequence of wild-type ACC SEQ ID NO: 2 Protein Amino acid sequence of I1879V SEQ ID NO: 3 Protein Amino acid sequence of C2186R SEQ ID NO: 4 Protein Amino acid sequence of I1879V and C2186R double-mutation SEQ ID NO: 5 DNA DNA sequence of I1879V SEQ ID NO: 6 DNA DNA sequence of C2186R SEQ ID NO: 7 DNA DNA sequence of I1879V and C2186R double-mutation

(151) The present disclosure has the following advantages:

(152) 1. The present disclosure screens out a group of mutant ACC proteins.

(153) 2. A plant with the mutant ACC protein of the present disclosure shows obvious herbicide resistance compared with the wild-type plant.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of an ABE-nCas9 base editor, where OsU6 and ZmUbi are promoters; sgRNA is a gRNA; bp-NLS is a nuclear localization signal; and NOS is a terminator.

(2) FIG. 2 shows the herbicide resistance of a plant with mutation of amino acid 1,879 of ACC relative to the wild-type.

(3) FIG. 3 shows the herbicide resistance of a plant with mutation of amino acid 2,186 of ACC relative to the wild-type.

(4) FIG. 4 shows the tolerance of an I1879V edited plant to sethoxydim.

(5) FIG. 5 shows the tolerance of an I1879V edited plant to phentodim after phentodim is sprayed for 20 d.

(6) FIG. 6 shows the tolerance of I1879V and C2186R edited plants to quizalofop-p and fenoxaprop-p-ethyl.

(7) FIG. 7 shows the tolerance of an I1879V edited plant to different herbicides.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(8) The present disclosure will be further explained below in conjunction with examples. The following examples are only preferred examples of the present disclosure, and are not intended to limit the present disclosure in other forms. Any technical personnel familiar with the profession may use the technical content disclosed above to derive equivalent examples through equivalent changes. Any simple modification or equivalent change made to the following examples according to the technical essence of the present disclosure without departing from the content of the solutions of the present disclosure shall fall within the protection scope of the present disclosure.

Example 1 Construction of a Gene Editing Vector and Screening of a Mutation Site

(9) 1. An ABE-nCas9 base editor targeting the *Oryza sativa* endogenous ACC gene was constructed.

(10) The ABE base editor could realize A/T->G/C base conversion within a specified sequence window. In the present disclosure, the ABE-nCas9 base editor was used as a vector, and a sgRNA (sgRNA shown in Table 1) was designed in the *Oryza sativa* endogenous ACC gene and cloned into the ABE-nCas9 vector to form a base editor targeting the *Oryza sativa* endogenous ACC gene. An amino acid sequence encoded by the *Oryza sativa* endogenous ACC gene was shown in SEQ ID No: 1.

(11) TABLE-US-00002 TABLE 1 sgRNA sequences (guide sequences) targeting

Oryza sativa ACC sgRNA No. guide-PAM sequence (5'-3') A-ACC1879

GAGAATATACATGGAAGTGC A-ACC2186 ATAGCACTCAATGCGGTCTG

(12) 2. *Oryza sativa* Genetic Transformation and Transgenic Plant Identification

(13) Multiple *Oryza sativa* varieties such as Wanzhijing 006, Nanjing 9108, and Jiahe 218 were used as experimental materials. The base editor constructed above was transformed into the *Oryza sativa* plants by *A. tumefaciens* to obtain gene-edited plants. Emerging seedlings were screened with a sethoxydim-containing medium at a concentration of SET-2: 2 mg/L. Or, edited seedlings were planted in a cultivation room. The edited seedlings and the control plants were sprayed with sethoxydim at a concentration of 0.5 g/L (corresponding to a field application dosage of 10 g. a.i./mu), and a survival rate of the seedlings was counted 10 d later.

(14) Through screening cultivation, herbicide-resistant plants were screened out on a sethoxydim-containing medium. As shown in FIG. 2, plants indicated by the arrow were able to tolerate the herbicide and showed resistance to the herbicide compared with other plants. The above herbicide-resistant plants were identified by PCR and sequencing, and it was found that the herbicide-resistant plants had an expected base substitution within a target range and a specific editing type was I1879V. In addition, herbicide-resistant plants were also screened out in the cultivation room. As shown in FIG. 3, edited plants were able to tolerate the herbicide and showed resistance to the herbicide compared with the control plants. The above herbicide-resistant plants were identified by PCR and sequencing, and it was found that the herbicide-resistant plants had an expected base substitution within a target range and a specific editing type was C2186R.

(15) For I1879V edited *Oryza sativa* plants, the dosage of sethoxydim was further increased to a corresponding field application dosage of 15 g. a.i./mu. 14 d later, the control was obviously yellow and dead, and the edited material grew normally, as shown in FIG. 4. The dosage of sethoxydim was further increased to 30 g. a.i./mu. After the plants were further cultivated for 20 d, all wild-type plants died, but the edited plants still showed significant herbicide resistance, with phytotoxicity only of about 30%.

Example 2 Resistance to Other Types of Herbicides

(16) The I1879V and C2186R edited *Oryza sativa* plants obtained in Example 1 were planted, and the resistance to other types of herbicides was tested, such as phentodim, quizalofop-p, fenoxaprop-p-ethyl, clethodim, haloxyfop-p-methyl, fenoxaprop, and butyrclofen.

(17) 2.1 Phentodim:

(18) Dibble *Oryza sativa* plants at the seedling stage in the field were selected and sprayed with phentodim at 20 g.a.i./mu (4 times a recommended dosage), and then results were observed. After the phentodim was sprayed for 7 d, the wild-type plants showed obvious chlorosis and spots, but the I1879V edited plants showed no visible phytotoxicity. After the phentodim was applied for 20 d, as shown in FIG. 5, the wild-type plants showed serious phytotoxicity (the arrow in FIG. 5 indicated obvious phytotoxicity), and its growth was also severely affected; and the I1879V edited plants showed little difference in biomass from the control group. That is, even 20 d later, the I1879V edited plants still showed excellent resistance to phentodim, had no obvious phytotoxicity, and were not affected in growth. However, compared with the wild-type control, the C2186R edited plant showed some resistance. Compared with the I1879V edited plant, the C2186R edited plant showed obvious phytotoxicity at 20 g.a.i./mu of phentodim, and there were chlorosis and spots at bases of leaves of the edited material, which was not safe enough for practical use.

(19) 2.2 Quizalofop-p and fenoxaprop-p-ethyl

(20) Quizalofop-p and fenoxaprop-p-ethyl were each applied at 5 g.a.i./mu to I1879V edited plants and C2186R edited plants, and results were observed 14 d later. As shown in FIG. 6, 14 d later, there was no significant difference between the edited plants and the wild-type control, indicating that the above-mentioned edited plants failed to produce tolerance to quizalofop-p or fenoxaprop-p-ethyl.

(21) 2.3 Clethodim, haloxyfop-p-methyl, fenoxaprop, and butyrclofen:

(22) The tolerance of the I1879V edited plants to clethodim, haloxyfop-p-methyl, fenoxaprop, and butyrclofen was tested, where the clethodim, haloxyfop-p-methyl, and fenoxaprop were each applied at concentrations of 5 g.a.i./mu (1×) and 10 g.a.i./mu (2×); and the butyrclofen was applied at concentrations of 10 g.a.i./mu (1×) and 20 g.a.i./mu (2×). After the herbicide was applied for 20 d, the growth of the edited plants was observed and the herbicide resistance of the edited plants was counted. As shown in FIG. 7, the I1879V edited plants showed some phytotoxicity at the above-mentioned 1× herbicide concentration, with a mortality rate of 30% to 50%; and at the above-mentioned 2× herbicide concentration, the edited plants were basically in a non-resistant state, with a mortality rate of 70% to 90%.

(23) All documents mentioned in the present disclosure are cited as references in the present application, as if each document was individually cited as a reference. In addition, it should be understood that various changes or modifications may be made to the present disclosure by those skilled in the art after reading the above teaching content of the present disclosure, and these equivalent forms also fall within the scope defined by the appended claims of the present disclosure.

Claims

1. A method for imparting phentodim resistance to a plant, comprising a step of allowing endogenous acetyl-CoA carboxylase (ACC) of the plant to mutate to introduce a mutant ACC into a plant cell, a plant seed, a plant tissue, a plant part, or the plant, wherein compared to an amino acid sequence of a parent ACC, the mutant ACC has a mutation at an amino acid corresponding to amino acid 1,879 of an amino acid sequence shown in SEQ ID NO: 1, wherein the amino acid 1,879 is mutated into valine (V); and the plant is rice; wherein a method for controlling weeds comprises: a) providing a phentodim-resistant plant prepared by the method for imparting the phentodim resistance to the plant; and b) applying an effective amount of phentodim to the phentodim-resistant plant and the weeds nearby to control the weeds near the phentodim-resistant plant.
 2. The method according to claim 1, wherein the parent ACC is derived from a monocotyledonous plant or a dicotyledonous plant.
 3. The method according to claim 2, wherein the parent ACC is derived from *Oryza sativa*.
 4. The method according to claim 1, wherein the method comprises a step of allowing the mutant ACC to express in the plant cell, the plant seed, the plant tissue, the plant part, or the plant.
 5. The method according to claim 1, wherein the method comprises: introducing the mutant ACC with a gene editing tool.
 6. The method according to claim 5, wherein the gene editing tool is one or more selected from the group consisting of clustered regularly interspaced short palindromic repeat (CRISPR), transcription activator-like effector nuclease (TALEN), and zinc-finger nuclease (ZFN).
 7. The method according to claim 2, wherein the method comprises a step of allowing the mutant ACC to express in the plant cell, the plant seed, the plant tissue, the plant part, or the plant.
 8. The method according to claim 3, wherein the method comprises a step of allowing the mutant ACC to express in the plant cell, the plant seed, the plant tissue, the plant part, or the plant.
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